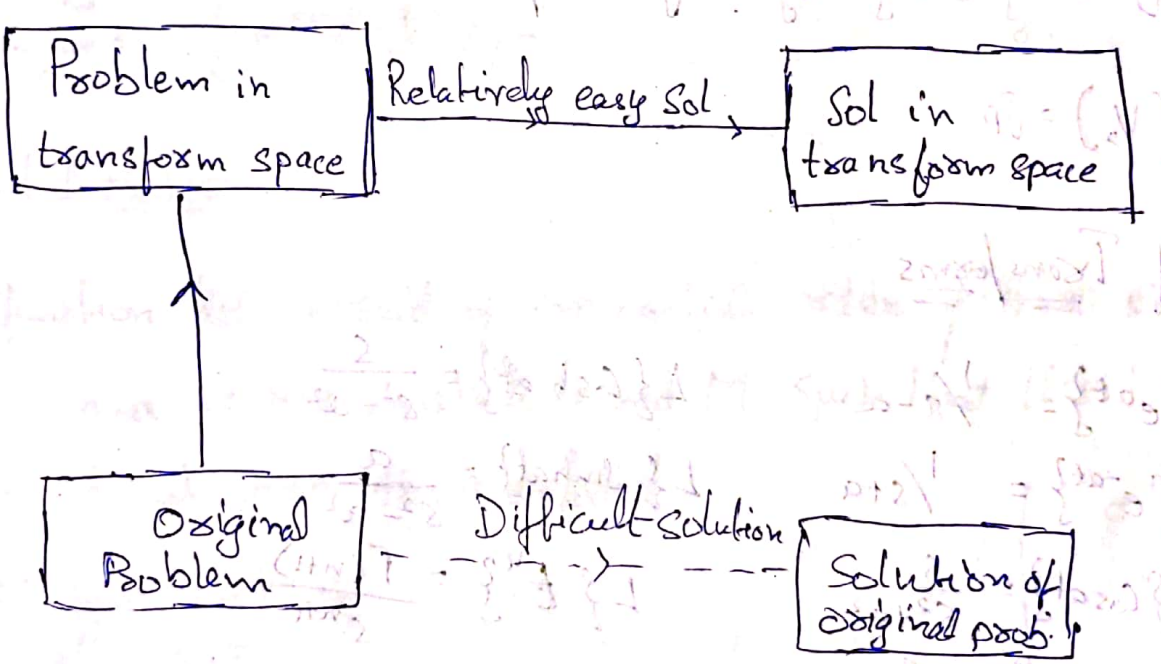
TT Sem Revision

Integral Transform

The transformation that involves integration in its process is called integral transform.

$$T[f(t)] = \int_a^b K(s,t)\, f(t)\, dt.$$

$K(s,t) \rightarrow$ Kernal.



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Laplace Transform.

$f(t)$ defined $t \geq 0$

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt.$$

Linear prop

$$L\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s).$$

→ $\Gamma(n) = \int_0^\infty e^{-y} y^{n-1} dy$. gamma function.

→ $\Gamma(1/2) = \sqrt{\pi}$

List of Transforms

→ $L\{e^{at}\} = 1/s-a$

→ $L\{e^{-at}\} = 1/s+a$

→ $L\{\cos at\} = \frac{s}{s^2+a^2}$

→ $L\{\sin at\} = \frac{a}{s^2+a^2}$

$L\{\cosh at\} = \frac{s}{s^2-a^2}$

$L\{\sinh at\} = \frac{a}{s^2-a^2}$

$L\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}$

$L\{k\} = \frac{k}{s}$.

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List of Inverse Transform

$\rightarrow L^{-1}[1/s-a] = e^{at}$

$\rightarrow L^{-1}[1/s+a] = e^{-at}$

$\rightarrow L^{-1}\left[\frac{s}{s^2+a^2}\right] = \cos at$

$\rightarrow L^{-1}\left[\frac{a}{s^2+a^2}\right] = \sin at$.

$\rightarrow L^{-1}\left[\frac{s}{s^2-a^2}\right] = \cosh at$

$\rightarrow L^{-1}\left[\frac{a}{s^2-a^2}\right] = \sinh at$

$\rightarrow L^{-1}\left[\frac{1}{s^n}\right] = \frac{t^{n-1}}{\Gamma(n)}$.

$\rightarrow L^{-1}\left[\frac{K}{s}\right] = K$.

→ Laplace Transform does not exist for all functions.

First Minimum Condⁿ $\int_0^\infty e^{-st} f(t)\,dt < \infty$.

Exponential order

A function $f(t)$ is said of exponential order if there exists a constant c and positive constants t_0 and M such that $|f(t)| < Me^{ct}$ for all $t > t_0$ at which $f(t)$ is defined.

Eo - t^2 Sint cost $1/t$

NEO - e^{t^2}, tant.

e^{t^2}, tant

log t

↓

no LT

Piecewise continuous function

A function f is piecewise cont. on an interval $t \in [a,b]$ if the interval can be partitioned by a finite number of points $a = t_0 < t_1 \ldots < t_n = b$ such that.

→ f is cont. on each open subinterval (t_{i-1}, t_i)

→ f approaches a finite limit as the endpoints of each subinterval are approached within the subinterval.

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Shifting Theorem

$\rightarrow L\{e^{at} f(t)\} = F(s-a)$

Transform of Derivative

$\rightarrow L\{f'(t)\} = sL\{f(t)\} - f(0)$.

$\rightarrow L\{f''(t)\} = s^2 L\{f(t)\} - s f(0) - f'(0)$.

Derivative of Transform

$\rightarrow L\{t f(t)\} = -\frac{d F(s)}{ds}$. $\rightarrow L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} L\{f(t)\}$

$\rightarrow L\{t^2 f(t)\} = \frac{d^2 F(s)}{ds}$

$\rightarrow L^{-1}[F'(s)] = -t L^{-1}[F(s)]$ [Qn with ln].

Transform of Integral

$\rightarrow L\{\int_0^t f(t)\, dt\} = \frac{F(s)}{s}$

$L^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t f(t)\, dt$.

$L^{-1}\left[\frac{F(s)}{s^2}\right] = \int_0^t \int_0^t f(t)\, dt$

Integration of Transform

$\rightarrow L\{\frac{f(t)}{t}\} = \int_s^\infty F(s)\, ds$. [Cond^n $\lim_{t \to 0^+} \frac{f(t)}{t}$ exists.

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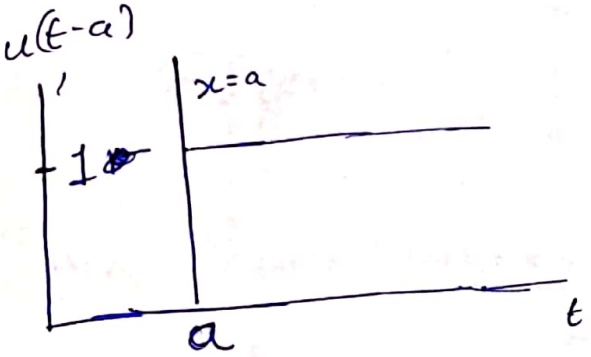
→ Laplace transform of $\frac{\cos t}{t}$ does not exist.

<u>Unit step or Heaviside function</u>

$$u(t-a) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t > a. \end{cases} \qquad [a \geq 0]$$



→ $L\{u(t-a)\} = \frac{e^{-as}}{s}$

1. $L\{f(t-a)u(t-a)\} = e^{-as}F(s)$.
2. $L\{f(t-a)\,u(t-a)\} = e^{-as}L\{f(t)\}$.

$$L\{f(t)\,u(t-a)\} = e^{-as}L\{f(t+a)\}.$$

3. $L^{-1}[e^{-as}F(s)] = f(t-a)\,u(t-a)$

$$= u(t-a)\,f(t-a)$$

$$= u(t-a)\,\{L^{-1}[F(s)]\}_{t \to (t-a)}.$$

$$i(t) = \frac{V_0}{R}\left\{ e^{-\frac{1}{RC}(t-a)}\,u(t-a) - e^{-\frac{1}{RC}(t-b)}\,u(t-b) \right\}.$$

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Unit Impulse (Dirac Delta) function

$$g(t-a) = \begin{cases} 1/h & a < t < a+h \\ 0 & \text{otherwise.} \end{cases}$$

$$\delta(t-a) = \lim_{h \to 0} g(t-a).$$

$\rightarrow L\{\delta(t-a)\} = e^{-as}.$

Convolution

$$f(t) * g(t) = \int_0^t f(u)\, g(t-u)\, du.$$

- Geometrically convolution means area of overlapping
- Physical meaning of convolution → Amount of overlapping
- Convolution is commutative, distributive, associative.

$$\int uv = uv_1 - u'v_2 + u''v_3 - \cdots$$

Convolution Theorem

1. $L\{f(t) * g(t)\} = F(s)\, G(s).$
2. $L^{-1}[F(s)\, G(s)] = L^{-1}[F(s)] * L^{-1}[G(s)].$

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Transform of periodic function

$$L\{f(t)\} = \frac{1}{1-e^{-ps}} \int_0^p e^{-st} f(t)\,dt.$$

Fourier Transform

$$F\{f(t)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ist} f(t)\,dt.$$

Inverse Fourier Transform

$$F^{-1}[F(s)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ist} F(s)\,ds.$$

Existence condition

absolutely integrable & piecewise continuous

$\hookrightarrow$ (a,b) $\int_a^b |f(t)|\,dt < \infty$

$$\rightarrow \int_0^{\infty} \frac{\sin x}{x}\,dx = \frac{\pi}{2} \qquad \left[f(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \cos st \frac{\sin s}{s}\,ds \right]$$

$$\Rightarrow F\{u(t-a)\} = \frac{1}{\sqrt{2\pi}} \left\{ \frac{e^{-ias}}{is} \right\}.$$

$$\rightarrow F\{\delta(t-a)\} = \frac{1}{\sqrt{2\pi}} e^{-ias}$$

$$\rightarrow f(t) = \frac{4}{\pi} \int_0^{\infty} \cos st \left(\frac{\sin s - s\cos s}{s^3} \right) ds.$$

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$\rightarrow F\{e^{-a^2x^2}\}$ $a>0$

$$= \frac{e^{\frac{-s^2}{4a^2}}}{a\sqrt{2}}$$ $a = 1/\sqrt{2}$ na self reciprocal.

<u>Fourier Sine</u>

$$F_s\{f(t)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \sin st \, dt$$

<u>Inverse formula</u>

$$F_s^{-1}\{F_s(s)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty F_s(s) \sin st \, ds$$

<u>Fourier Cosine</u>

$$F_c\{f(t)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \cos st \, dt.$$

<u>Inverse</u>

$$F_c\{f(t)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty F_c(s) \cos st \, ds.$$

$$F_s\{e^{-ax}\} = \sqrt{\frac{2}{\pi}} \frac{s}{s^2+a^2}$$

$$F_c\{e^{-ax}\} = \sqrt{\frac{2}{\pi}} \frac{a}{s^2+a^2}$$

$\rightarrow$ If $f(x)$ is even function then $F_c\{f(x)\} = F\{f(x)\}$.

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$\rightarrow F_c\{x^{n-1}\} = \sqrt{\frac{2}{\pi}}\left\{\frac{\cos(\frac{n\pi}{2})}{s^n}\right\}\Gamma(n)$

$\rightarrow F_s\{x^{n-1}\} = \sqrt{\frac{2}{\pi}}\left\{\frac{\sin(\frac{n\pi}{2})}{s^n}\right\}\Gamma(n)$

$\rightarrow F_c\left\{\frac{1}{x^2+1}\right\} = \sqrt{\frac{\pi}{2}}\, e^{-s}$

Linear prop

$F\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$

Change of Scale.

$F\{f(at)\} = \frac{1}{|a|} F(s/a).$

S-shifting.

$F\{e^{-iat} f(t)\} = F(s+a)$

t-shifting

$F\{f(t-a)\} = e^{-ias} F(s).$

Modulation Prop

$F_s\{f(t)\sin at\} = \frac{-1}{2}[F_c(s+a) - F_c(s-a)]$

$F\{f(t)\cos at\} = \frac{1}{2}[F(s-a) + F(s+a)]$

$F_c\{f(t)\cos at\} = \frac{1}{2}[F_c(s+a) + F_c(s-a)]$

$F_c\{f(t)\sin at\} = \frac{1}{2}[F_s(s+a) - F_s(s-a)]$

$F_s\{f(t)\cos at\} = \frac{1}{2}[F_s(s+a) + F_s(s-a)]$

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Transform of derivative

If $f(t)$ is continuous $f'(t)$ is piecewise continuously diff $f(t)$, $f'(t)$ are absolutely integrable in $(-\infty, \infty)$ and $\lim_{t \to \pm\infty} f(t)$

then $F\{f'(t)\} = is F(s)$.

Derivative of Transform

If $F\{f(t)\} = F(s)$ then $F\{t f(t)\} = i \frac{d}{ds} F(s)$.

Faultang theorem.

$$F\{f(t) * g(t)\} = F(s) G(s).$$

Parseval's Identity (or) Energy theorem.

$F\{f(t)\} = F(s)$ then $\int_{-\infty}^{\infty} |f(t)|^2 dt = \int_{-\infty}^{\infty} |F(s)|^2 ds$.

i) $\int_0^{\infty} f(t) g(t) dt = \int_0^{\infty} F_c(s) G_c(s) ds = \int_0^{\infty} F_s(s) G_s(s) ds$.

ii) $\int_0^{\infty} |f(t)|^2 dt = \int_0^{\infty} |F_s(s)|^2 ds = \int_0^{\infty} |F_c(s)|^2 ds$.

$\rightarrow \int_{-\infty}^{\infty} \frac{\sin^2 ax}{x^2} = a\pi$

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$$\rightarrow \int_0^\infty \frac{dx}{(a^2+x^2)^2} = \frac{\pi}{4a^3}$$

$$\rightarrow \int_0^\infty \frac{x^2\,dx}{(a^2+x^2)^2} = \frac{\pi}{4a}$$

$$\rightarrow \int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2(a+b)ab}.$$

$$\rightarrow \int_0^\infty \frac{x^2\,dx}{(a^2+x^2)(x^2+b^2)} = \frac{\pi}{2(a+b)}.$$

$$\rightarrow F_s\{x f(x)\} = -\frac{d}{ds} F_c\{f(x)\}.$$

$$\rightarrow F_c\{x f(x)\} = \frac{d}{ds} F_s\{f(x)\}.$$

$$\rightarrow F\{x e^{-ax}\} = \sqrt{\frac{2}{\pi}}\left\{\frac{(a^2-s^2)}{(a^2+s^2)^2}\right\}.$$

$$\rightarrow F\{x e^{-a^2x^2}\} = +\frac{2s}{4a^3\sqrt{2}} e^{-\frac{s^2}{4a^2}}$$

Fourier Integral Formula

$$f_L(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos\omega_n x + b_n \sin\omega_n x) \quad \omega_n = \frac{n\pi}{L}$$

$$f(x) = \int_0^\infty [A(\omega)\cos\omega x + B(\omega)\sin\omega x]\,d\omega.$$

where

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{+\infty} f(v)\cos\omega v\,dv$$

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v\,dv.$$

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If $f(x)$ is even function in $(-\infty, \infty)$

$$f(x) = \int_0^\infty A(w) \cos wx \, dw.$$ Cosine Integral

where $$A(w) = \frac{2}{\pi} \int_0^\infty f(v) \cos wv \, dv.$$

If $f(x)$ is an odd function in $(-\infty, \infty)$ then the Fourier Integ becomes. Sine Integral.

$$f(x) = \int_0^\infty B(w) \sin wv \, dv.$$

Discrete Convolution

$x(n)$ - Duration N_1

$y(n)$ - Duration N_2

$$z(n) = x(n) * y(n) = \sum_{r=0}^{n} x(r) y(n-r) \rightarrow \text{Duration } N_1+$$

Periodic Sequence Period N $x(n+N) = x(n)$.

Matrix Multiplication Method (MMM)

$$z = \begin{bmatrix} z(0) \\ z(1) \\ z(2) \\ z(3) \end{bmatrix} \quad y = \begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix}$$

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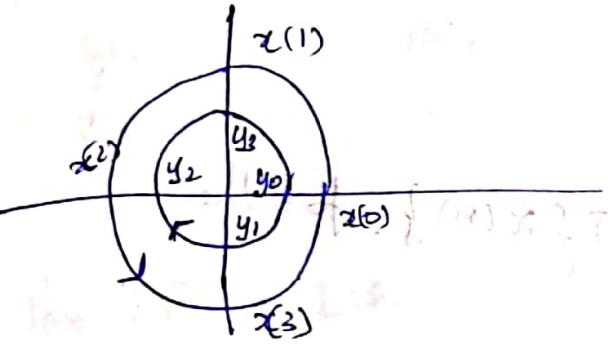
$$X = \begin{bmatrix} x(0) & x(3) & x(2) & x(1) \\ x(1) & x(0) & x(3) & x(2) \\ x(2) & x(1) & x(0) & x(3) \\ x(3) & x(2) & x(1) & x(0) \end{bmatrix}$$

$$Z = XY.$$

Circular Representation Method [CRM].



Rotate index by one in anticlockwise.

Discrete Fourier Transform

Suppose $\{x(n)\}$ is assumed to be a finite sequence of duration N then the DFT of $\{x(n)\}$ denoted as $DFT\{x(n)\}$

$$DFT\{x(n)\} = \sum_{n=0}^{N-1} x(n)\, e^{-\frac{2\pi i n k}{N}}$$

$$DFT\{x(n)\} = X(k) \quad \text{for } K = 0, 1, \ldots N-1$$

Phase factor

$$W_N = e^{-\frac{2\pi i}{N}}$$

$$DFT\{x(n)\} = \sum_{n=0}^{N-1} x(n)\, W_N^{nk}$$

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Inverse formula

$$IDFT\{X(k)\} = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{\frac{2\pi i n k}{N}}$$

$$IDFT\{X(k)\} = x(n) \quad \text{for } n = 0, 1, \ldots N-1$$

$$x(n) = IDFT\{X(k)\} = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, W_N^{-nk}$$

→ It is linear.

Shifting property

$$DFT\{x(n-m)\} = W_N^{mk}\, DFT\{x(n)\}$$ for the periodic sequence of N.

Convolution

$$DFT\{x(n) * y(n)\} = DFT\{x(n)\}\, DFT\{y(n)\}.$$

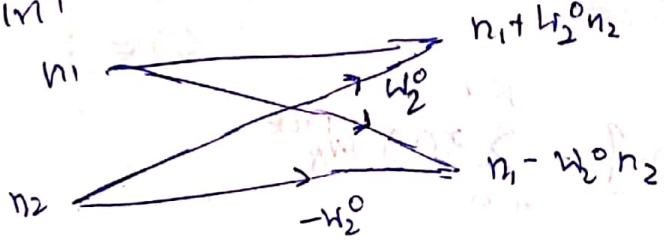
Fast Fourier Transform [FFT] [only radix 2].

DFT takes N^2 computation

FFT takes $N \log N$.

DIT.

2point



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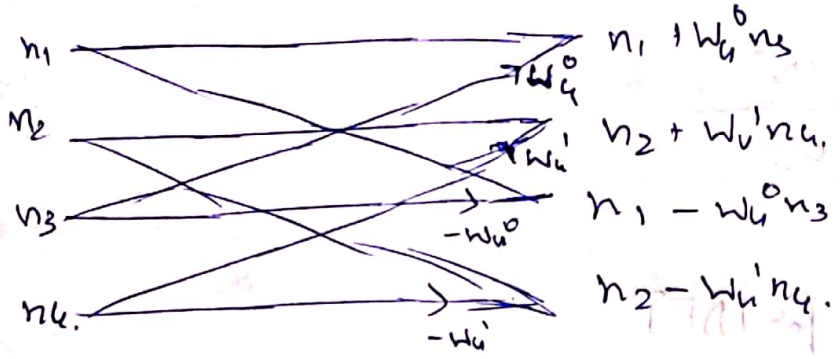
4point.



$W_2 = -1$

$W_4 = -i$

$W_8 = \frac{1-i}{\sqrt{2}}$ $\quad W_8^2 = -i \quad W_8^3 = \frac{-1-i}{\sqrt{2}}$

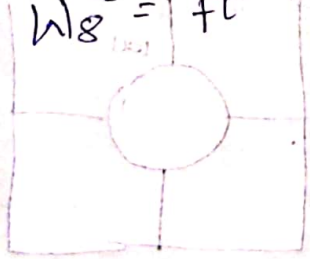
For IDFT in DIT we have 2 changes.

→ Power of phase factor in negative

→ After getting final ans divide by duration.

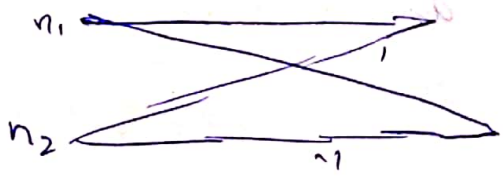
$W_4^{-1} = i$

$W_8^{-1} = \frac{1+i}{\sqrt{2}}$ $\quad W_8^{-2} = +i \quad W_8^{-3} = \frac{-1+i}{\sqrt{2}}$



For DIF

2point.



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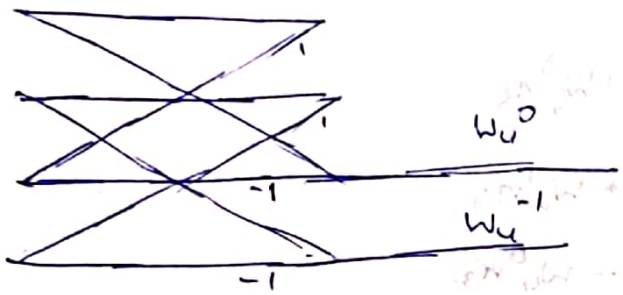


Same for DIF as in DIT for IDFT

Z Transform

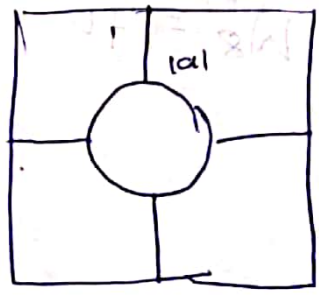
Suppose $\{x(n)\}$ is a sequence of infinite duration then the Z Transform of $x(n)$ denoted as $Z\{x(n)\}$ and defined as

$$Z\{x(n)\} = \sum_{n=0}^{\infty} x(n) z^{-n}$$ [z is a complex variable].

$$Z\{x(n)\} = X(z)$$ → function of complex num
↳ takes values in a plane.

→ Helps to transform from Discrete to cont.

$$Z\{a^n\} = \frac{z}{z-a}$$



Exterior of a circle of radius a.

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Table of Transform

$$Z\{a^n\} = \frac{z}{z-a} \quad |a|<|z|$$

$$Z\{\frac{1}{n}\} = \log\frac{z}{z-1} \quad |z|>1$$

$$Z\{n\} = \frac{z}{(z-1)^2} \quad |z|>1$$

$$Z\{a^n n\} = \frac{az}{(z-a)^2}$$

$$Z\{n+1\} = \frac{z^2}{(z-1)^2}$$

$$Z\{1\} = \frac{z}{z-1}$$

$$Z\{k\} = \frac{kz}{z-1}$$

$$Z\{(\cos\frac{n\pi}{2})\} = \frac{z^2}{z^2+1}$$

$$Z\{(\sin\frac{n\pi}{2})\} = \frac{z}{z^2+1}$$

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots$$

$$Z(r^n(\cos n\theta)) = \frac{z^2 - zr\cos\theta}{z^2 - 2zr\cos\theta + r^2}$$

$$Z(r^n(\sin n\theta)) = \frac{zr\sin\theta}{z^2 - 2zr\cos\theta + r^2}$$

Table of Inverse

$$Z^{-1}\left[\frac{z}{z-a}\right] = a^n$$

$$Z^{-1}\left[\log\frac{z}{z-1}\right] = \frac{1}{n}$$

$$Z^{-1}\left[\frac{z}{(z-1)^2}\right] = n$$

$$Z^{-1}\left[\frac{az}{(z-a)^2}\right] = a^n n$$

$$Z^{-1}\left[\frac{z^2}{(z-1)^2}\right] = n+1$$

$$Z^{-1}\left[\frac{z^2}{z^2+1}\right] = \cos\left(\frac{n\pi}{2}\right)$$

$$Z^{-1}\left[\frac{z}{z^2+1}\right] = \sin\left(\frac{n\pi}{2}\right)$$

$$Z^{-1}\left[\frac{z}{z-1}\right] = 1$$

$$Z^{-1}\left[\frac{kz}{z-1}\right] = k$$

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-> z is linear.

- Change of Scale prop

$$Z\{a^n x(n)\} = X(z/a)$$

Shifting prop

$$Z\{x(n-k)\} = z^{-k} Z\{x(n)\}.$$

$$Z\{x(n+1)\} = zZ\{x(n)\} - z\,x(0).$$

$$Z\{x(n+2)\} = z^2 Z\{x(n)\} - z^2 x(0) - z\,x(1).$$

$$Z\{x(n+3)\} = z^3 Z\{x(n)\} - z^3 x(0) - z^2 x(1) - z\,x(2).$$

Derivative of Transform

$$Z\{n\,x(n)\} = -z \frac{d}{dz} X(z).$$

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